<table>
<tr><th colspan="3" rowspan="2">Question</th><th rowspan="2">Marking details</th><th colspan="6">Marks available</th></tr>
<tr><th>AO1</th><th>AO2</th><th>AO3</th><th>Total</th><th>Maths</th><th>Prac</th></tr>
<tr><td>6</td><td>(a)</td><td></td><td>2 materials chosen and microscopic structure of each explained [2 × (1)] See below.
1 example given of each material [2 × (1)] See below.
Crystalline - long range order / lattice like arrangement/regular arrangement structure e.g. metals
Amorphous - short range order / irregular or random arrangement / no order e.g. glass, ceramics, brick
Polymeric - long chain molecule arrangement [of hydrocarbons] e.g. rubber, polythene accept plastic</td><td>4</td><td></td><td></td><td>4</td><td></td><td></td></tr>
<tr><td></td><td>(b)</td><td></td><td>Indicative content:
Measurements:
• Extension of wire [with pointer/ruler]
• Original length of wire [from clamp to pointer]
• Diameter of wire using micrometer or (Vernier) callipers

Determination of Young modulus:
• Use $E = \frac{Fl}{Ae}$ or plot graph of load/extension or stress/strain and find gradient
• A determined from $\pi\left(\frac{d}{2}\right)^2$

Precautions:
• Repeat readings of e (adding/removing load)
• Measure {d in various places on wire / mean d}
• Keep temperature constant / use of Searle's apparatus
• Ensure no kinks in wire
• Soft wood so as not to damage wire
• Ensure wire is securely clamped
• Stay within elastic limit
• Use of a longer wire / travelling miroscope
• Avoid parallax</td><td>4</td><td></td><td>2</td><td>6</td><td></td><td>6</td></tr>
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Question			Marking details	AO1	AO2	AO3	Total	Maths	Prac
				Marks available					
			5-6 marks Comprehensive account with reference to how the measurements must be made, how they should be used to determine Young modulus and precautions that should be taken to minimise uncertainties. *There is a sustained line of reasoning which is coherent, relevant, substantiated and logically structured.* **3-4 marks** Comprehensive account with reference to 2 out of 3 of the following or a limited attempt at all 3 areas - how the measurements must be made, how they should be used to determine Young modulus and precautions that should be taken to minimise uncertainties. *There is a line of reasoning which is partially coherent, largely relevant, supported by some evidence and with some structure.* **1-2 marks** Comprehensive account with reference to 1 out of 3 of the following or a limited attempt at 2 areas - how the measurements must be made, how they should be used to determine Young modulus and precautions that should be taken to minimise uncertainties. *There is a basic line of reasoning which is not coherent, largely irrelevant, supported by limited evidence and with very little structure.* **0 marks** No attempt made or no response worthy of credit.						
			Question 6 total	8	0	2	10	0	6